Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ata set has a mean of 10. An outlier of 100 is added to the high end. Does the mean increase? Student says:

Think about how one large value changes the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outlier raises the total, so the mean increases.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which measure of center is resistant to outliers: mean or median? Student's answer: mean

Think about which measure depends on the middle value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The median depends only on the middle values, so it is resistant. The mean is pulled by outliers.	**Median**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The data 1 through 9 gets an outlier of 100 added. What is the new range? Student's answer: 100

Subtract the smallest value from the largest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The range is 100 - 1 = 99. The student gave the maximum.	**99**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If an outlier is removed from a data set, does the standard deviation usually increase? Student says: Yes

Think about how far the outlier was from the mean.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outlier was far from the mean, so removing it usually decreases the standard deviation.	**No, it usually decreases**

5

Find the Error

S.ID.3

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ved right with a few large values. Is the mean greater than or less than the median? Student's answer: The r

Think about how the large values pull the mean.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Large values pull the mean to the right, so the mean is greater than the median.	**The mean is greater**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

data 50, 52, and 54 has the median 52. An outlier of 0 is added. What is the new median? Student's answe

Order all four values and find the middle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
With values 0, 50, 52, 54, the median is (50 + 52) ÷ 2 = 51.	**51**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

n has mean 100 and standard deviation 15. About 68% of values lie between which two numbers? Student's

Use the empirical rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
70 to 130 is within 2 standard deviations, about 95%. Within 1 standard deviation, the values are 85 and 115.	**85 and 115**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

With mean 100 and standard deviation 15, what is the z-score of 130? Student's answer: 30

Use z = (x - mean) ÷ standard deviation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not divide by the standard deviation. z = (130 - 100) ÷ 15 = 2.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

distribution, about what percent of values lie more than 1 standard deviation above the mean? Student's a

Use the empirical rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Half of values are above the mean, but only about 16% are more than 1 standard deviation above it.	**About 16%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

what percent of values in a normal distribution lie within 2 standard deviations of the mean? Student's answ

Recall the empirical rule for 2 standard deviations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
About 68% is within 1 standard deviation. About 95% is within 2 standard deviations.	**About 95%**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ercent of values in a normal distribution lie between the mean and 1 standard deviation above it? Student's

Think about half of the 68% band.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 68% band covers 1 standard deviation on each side. Half of it, about 34%, lies between the mean and 1 standard deviation above.	**About 34%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ormal distribution has mean 50 and standard deviation 8. What value has a z-score of -1? Student's answe

Use x = mean + z × standard deviation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added. x = 50 + (-1)(8) = 42.	**42**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

y of 100 students is taken to learn about the whole school. What is the whole school? Student's answer: the

Decide which group is being studied.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 100 students are the sample. The whole school is the population.	**The population**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The mean of all voters' opinions is a parameter. Is it a parameter or a statistic? Student's answer: statistic

Parameters describe populations; statistics describe samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The mean of all voters describes the whole population, so it is a parameter.	**Parameter**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A statistic is a number computed from which group? Student's answer: the population

Think about where statistics come from.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A statistic is computed from a sample. A parameter is computed from the population.	**A sample**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a census a sample? Student says: Yes

Think about whether everyone is included.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A census includes every member of the population, so it is not a sample.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

an is used to estimate the population mean. Is the sample mean a parameter or a statistic? Student's answ

Decide whether it describes a sample or population.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sample mean is computed from a sample, so it is a statistic.	**Statistic**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

portion of all 10,000 students who like math is a number. Is it a parameter or a statistic? Student's answer: s

Decide whether the group is the whole population.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
All 10,000 students is the whole population, so the proportion is a parameter.	**Parameter**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

el says a coin lands heads 10% of the time. In 50 flips, there are 20 heads. Is the model plausible? Student sa

Compare the result to what the model predicts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The model predicts about 5 heads in 50 flips, but 20 is far from that. The model is not plausible.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A model says a die is fair. In 60 rolls, a 6 comes up 30 times. Is the model plausible? Student says: Yes

Compare the result to what the model predicts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A fair die gives 6 about 10 times in 60 rolls. 30 is far from the prediction, so the model is not plausible.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

model says 50% of students like pizza. In a sample of 20, 11 like pizza. Is the model plausible? Student says: N

Compare the result to what the model predicts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The model predicts about 10 of 20, and 11 is close. The model is plausible.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

del says the chance of success is 25%. In 40 trials there are 8 successes. Is the model plausible? Student sa

Compare the result to what the model predicts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The model predicts 10 successes in 40 trials, and 8 is close. The model is plausible.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

a result considered unusual if it is far from what a model predicts? Student says: No, results are always usu

Think about what a model predicts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A result far from the model's prediction is unusual and gives evidence against the model.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ir coin, the chance of 0 heads in 10 flips is about 0.1%. If this happens, is the coin model plausible? Student s

Think about how unlikely the result is.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
An event with a 0.1% chance is very unlikely under a fair coin model, so the model is not plausible.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A survey uses only people who choose to answer an online poll. Is this sample biased? Student says: No

Think about who is likely to respond.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
People who choose to respond may differ from the population, so the sample is biased.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A survey asks: Don't you agree that homework is awful? Is the question biased? Student says: No

Look at the wording.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The wording pushes respondents toward one answer, so the question is biased.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e of 100 is selected from a population. Is the sample mean an unbiased estimate of the population mean?

Think about what random selection does.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Random sampling gives an unbiased estimate, so the sample mean is unbiased.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ects every 10th student from an alphabetical list. What kind of sample is this? Student's answer: simple ran

Think about how the students are chosen.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Choosing every 10th person is a systematic sample, not a simple random sample.	**Systematic sample**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ses only students in one class to learn about the whole school. Is this sample biased for the school? Studer

Think about who is included.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
One class is not representative of the whole school, so the sample is biased.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is a stratified sample? Student's answer: a random sample from only one group

Think about groups within the population.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A stratified sample divides the population into groups and samples from each group in proportion to its size.	**A sample taken in proportion from each group**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

dom sample of 200 people has 120 who support a plan. Estimate the population proportion. Student's answ

Find the proportion in the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the count. The proportion is 120/200 = 60%.	**60%**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

l has a sample proportion of 60% and a margin of error of ±3%. What is the interval? Student's answer: 3% to

Subtract and add the margin of error.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted incorrectly. The interval is 57% to 63%.	**57% to 63%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sample mean is 40 with a margin of error of ±2. What is the interval? Student's answer: 40 to 42

Subtract and add the margin of error.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added only to the mean. The interval is 38 to 42.	**38 to 42**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Does a larger sample size give a smaller margin of error? Student says: No

Think about how sample size affects precision.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Larger samples give more precise estimates, so the margin of error is smaller.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e has 55% support and a margin of error of ±4%. What is the plausible range for the population? Student's

Find the lower and upper bounds.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student only added the margin. The range is 51% to 59%.	**51% to 59%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 95% confidence level means that: 95% of the population is in the interval. Student says this is correct.

Think about what the confidence level describes.

What did the student do wrong? Why do you think the student made this error?

A 95% confidence level means that about 95% of samples would produce intervals that capture the true parameter, not that 95% of the population is in the interval.

Solve the problem correctly. Show all of your work.

Incorrect

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

re subjects randomly assigned to treatment and control groups? Student's answer: to make the groups dif

Think about what random assignment balances.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Random assignment balances other factors between the groups, so differences can be attributed to the treatment.	**To balance other factors between groups**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

xperiment finds a treatment causes improvement. Can we conclude the treatment caused the change? S

Think about what randomized experiments allow.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Randomized experiments allow causal conclusions when the difference is significant.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

reakfast have higher grades in an observational study. Can we conclude breakfast causes higher grades?

Think about whether the study was randomized.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Observational studies show association, not cause. Other factors may explain the difference.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ows a 5-point difference happens by chance only 2% of the time. Is the difference statistically significant? S

Compare the chance to a common cutoff.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A 2% chance is small, which suggests the difference is statistically significant.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the purpose of a placebo group in an experiment? Student's answer: to increase the sample size

Think about what the control is comparing against.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A placebo group helps separate the effect of the treatment from the effect of expecting a treatment.	**To separate the treatment effect from expectations**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In a blind experiment, do participants know which treatment they receive? Student says: Yes

Think about what blinding means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
In a blind experiment, participants do not know their group, which reduces bias.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

dy finds coffee drinkers have higher risk. A headline says coffee causes the risk. Is this conclusion supporte

Think about observational studies.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Observational studies show association, not cause, so the headline overstates the finding.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

0 teachers in one school is reported as showing what all teachers think. Is this a valid generalization? Stud

Think about who was surveyed.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Fifty teachers from one school do not represent all teachers, so the generalization is not valid.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ndom sample of 1,000 or a random sample of 10 gives an estimate. Which is more reliable? Student's answe

Think about how sample size affects reliability.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Larger random samples give more reliable estimates. The sample of 1,000 is more reliable.	**1,000**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A study of 3 patients reports a cure rate of 100%. Is this result reliable? Student says: Yes

Think about the size of the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A sample of 3 is too small to give a reliable estimate.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can results from volunteers in one city be generalized to the whole country? Student says: Yes

Think about who was sampled.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Results apply to the population sampled. Volunteers from one city do not represent the whole country.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A report does not say how the sample was chosen. Is the report's claim fully reliable? Student says: Yes

Think about what information is needed to judge a sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Without knowing how the sample was chosen, we cannot judge whether it is representative.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many outcomes are in the sample space for flipping two coins? Student's answer: 2

List all possible outcomes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outcomes are HH, HT, TH, and TT, so there are 4.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many outcomes are in the event 'at least one head' when flipping two coins? Student's answer: 1

List outcomes with at least one head.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outcomes are HH, HT, and TH, so there are 3.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

On a standard die, what is the complement of the event 'rolling an even number'? Student's answer: {2, 4, 6}

Find the outcomes not in the event.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student listed the even numbers again. The complement is {1, 3, 5}.	**{1, 3, 5}**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A = {1, 2, 3} and B = {3, 4}. How many outcomes are in A or B? Student's answer: 5

Combine the lists without repeating.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The union is {1, 2, 3, 4}, which has 4 outcomes. The 3 is counted only once.	**4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A = {2, 4, 6} and B = {4, 5}. What is A and B? Student's answer: {4, 5}

Find the outcomes in both sets.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student listed B. The only outcome in both sets is 4, so A and B = {4}.	**{4}**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many ways can two dice have a sum of 2? Student's answer: 2

List the pairs that give a sum of 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Only (1, 1) gives a sum of 2, so there is 1 way.	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are the events 'first flip is heads' and 'second flip is heads' independent? Student says: No

Check whether one event changes the other's probability.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outcome of the first flip does not change the second flip, so the events are independent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. Are A and B independent? Student says: No

Check whether $P(A \text{ and } B) = P(A) \times P(B)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$0.5 \times 0.4 = 0.2$, so the events are independent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.3$, $P(B) = 0.6$, and $P(A \text{ and } B) = 0.3$. Are A and B independent? Student says: Yes

Check whether $P(A \text{ and } B) = P(A) \times P(B)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$0.3 \times 0.6 = 0.18$, which is not 0.3. The events are not independent.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two cards are drawn from a deck without replacement. Are the draws independent? Student says: Yes

Think about whether the first draw changes the deck.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Without replacement, the first card changes the second draw, so the draws are not independent.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

P(A) = 1/3 and P(B) = 1/4, and A and B are independent. What is P(A and B)? Student's answer: 7/12

Multiply the probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the probabilities. For independent events, P(A and B) = 1/3 × 1/4 = 1/12.	**1/12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A die is rolled. What is P(even and greater than 3)? Student's answer: 1/2

List outcomes in both events.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outcomes that are even and greater than 3 are {4, 6}, so the probability is 2/6 = 1/3.	**1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

P(A and B) = 0.2 and P(B) = 0.5. What is P(A | B)? Student's answer: 0.2

Use P(A | B) = P(A and B) ÷ P(B).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student repeated P(A and B). P(A | B) = 0.2 ÷ 0.5 = 0.4.	**0.4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 18 girls and 12 boys. 6 girls wear glasses. What is P(glasses | girl)? Student's answer: 1/5

Use only the girls in the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used all 30 students. The condition is a girl, so P = 6/18 = 1/3.	**1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.4$ and $P(A \mid B) = 0.4$. Are A and B independent? Student says: No

Compare $P(A \mid B)$ with $P(A)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$P(A \mid B) = P(A)$, so the events are independent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A card is drawn from a standard deck. What is P(king | face card)? Student's answer: 4/52

Use only face cards in the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the whole deck. There are 12 face cards and 4 kings, so the probability is 4/12 = 1/3.	**1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A \text{ and } B) = 0.3$ and $P(B) = 0.6$. What is $P(A \mid B)$? Student's answer: 0.3

Divide P(A and B) by P(B).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not divide. 0.3 ÷ 0.6 = 0.5.	**0.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A class has 20 boys, and 12 of them like sports. What is P(likes sports | boy)? Student's answer: 12/30

Use only the boys in the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 30 instead of 20. P = 12/20 = 0.6.	**0.6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

survey of 100 people, 40 have trait A, 30 have trait B, and 15 have both. What is P(A or B)? Student's answer:

Use P(A or B) = P(A) + P(B) − P(A and B).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not subtract the overlap. P(A or B) = (40 + 30 − 15) ÷ 100 = 0.55.	**0.55**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Using the same survey, what is P(A and B)? Student's answer: 0.40

Find the people with both traits.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the count for trait A. Both traits are 15 out of 100, so P = 0.15.	**0.15**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Using the same survey, what is P(B | A)? Student's answer: 0.15

Divide the overlap by the group with trait A.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave P(A and B). P(B | A) = 15 ÷ 40 = 0.375.	**0.375**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.4$, $P(B) = 0.3$, and $P(A \text{ and } B) = 0.15$. Are A and B independent? Student says: Yes

Check whether $P(A \text{ and } B) = P(A) \times P(B)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$0.4 \times 0.3 = 0.12$, which is not 0.15. The events are not independent.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

P(A) = 0.4. What is P(not A)? Student's answer: 0.4

Use the complement rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
P(not A) = 1 - 0.4 = 0.6.	**0.6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Of 80 students, 50 wear glasses. What is P(glasses)? Student's answer: 3/8

Write the count with glasses over the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the number without glasses. P = 50/80 = 5/8.	**5/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.2$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.1$. Are A and B independent? Student says: No

Check whether $P(A \text{ and } B) = P(A) \times P(B)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$0.2 \times 0.5 = 0.1$, so the events are independent.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

vents A and B are mutually exclusive, with P(A) = 0.3 and P(B) = 0.4. Are they independent? Student says: Ye

Find P(A and B) for mutually exclusive events.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Mutually exclusive events have P(A and B) = 0, but 0.3 × 0.4 = 0.12. They are not independent.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If $P(A \mid B) = P(A)$, are A and B independent? Student says: No

Think about what independence means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Independence means knowing B does not change the probability of A, which is exactly $P(A \mid B) = P(A)$.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fair coin is flipped 3 times. What is the probability of 3 heads? Student's answer: 1/6

Multiply the probabilities for independent flips.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided instead of multiplying. $(1/2)^3 = 1/8$.	**1/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are draws made with replacement independent? Student says: No

Think about whether the first draw changes the second.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
With replacement, the deck is the same for each draw, so the draws are independent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

P(rain | cloudy) = 0.6 and P(rain) = 0.6. Are rain and cloudy independent? Student says: No

Compare P(rain | cloudy) with P(rain).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The two probabilities are equal, so the events are independent.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

P(A and B) = 0.05 and P(B) = 0.25. What is P(A | B)? Student's answer: 0.05

Use P(A | B) = P(A and B) ÷ P(B).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student repeated P(A and B). P(A | B) = 0.05 ÷ 0.25 = 0.2.	**0.2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ms and 5 are defective. If the first item is defective and not replaced, what is P(second is defective)? Studer

Remove the first item from the count.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the original probability. After one defective item is removed, 4 defective items remain out of 49, so the probability is 4/49.	**4/49**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 40 seniors, and 15 of them are in honors. What is P(honors | senior)? Student's answer: 15/60

Use only seniors in the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 60 instead of 40. P = 15/40 = 3/8.	**3/8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(B \mid A) = 0.6$ and $P(A) = 0.5$. What is $P(A \text{ and } B)$? Student's answer: 0.6

Multiply P(A) by P(B | A).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave P(B | A). P(A and B) = 0.5 × 0.6 = 0.3.	**0.3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

There are 10 squares, and 4 of them are red. What is P(red | square)? Student's answer: 4/20

Use the squares in the denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 20 instead of 10. P = 4/10 = 0.4.	**0.4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If every outcome in event B is also in event A, what is P(A | B)? Student's answer: 0

Think about whether A always happens when B does.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
If B happens, then A must also happen, so P(A | B) = 1.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.5$, $P(B) = 0.4$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$? Student's answer: 0.9

Use $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not subtract the overlap. $P(A \text{ or } B) = 0.5 + 0.4 - 0.2 = 0.7$.	**0.7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A and B are mutually exclusive, with $P(A) = 0.2$ and $P(B) = 0.3$. What is $P(A \text{ or } B)$? Student's answer: 0.06

Add the probabilities for mutually exclusive events.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied. For mutually exclusive events, P(A or B) = 0.2 + 0.3 = 0.5.	**0.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A) = 0.6$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.3$. What is $P(A \text{ or } B)$? Student's answer: 1.1

Use the addition rule with the overlap.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not subtract the overlap. **$P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$.**	**0.8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$P(A \text{ or } B) = 0.9$, $P(A) = 0.6$, and $P(B) = 0.5$. What is $P(A \text{ and } B)$? Student's answer: 0.3

Rearrange the addition rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$P(A \text{ and } B) = P(A) + P(B) - P(A \text{ or } B) = 0.6 + 0.5 - 0.9 = 0.2$.	**0.2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is P(A or B) always less than or equal to 1? Student says: No

Think about the range of probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Every probability is between 0 and 1, so P(A or B) is always at most 1.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A and B are independent, with P(A) = 0.5 and P(B) = 0.5. What is P(A or B)? Student's answer: 1

Find P(A and B) first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
P(A and B) = 0.25. Then P(A or B) = 0.5 + 0.5 - 0.25 = 0.75.	**0.75**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e pays $10 with probability 1/5 and costs $2 with probability 4/5. What is the expected value? Student's answ

Multiply each outcome by its probability and add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the loss instead of subtracting it. Expected value = 10(1/5) - 2(4/5) = 2 - 1.6 = 0.4.	**0.4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the expected value of rolling a fair six-sided die? Student's answer: 3

Add the faces and divide by 6.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The average of 1 through 6 is 21 ÷ 6 = 3.5.	**3.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

random variable X takes values 0, 1, and 2 with probabilities 0.5, 0.3, and 0.2. What is E(X)? Student's answer

Multiply each value by its probability and add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
E(X) = 0(0.5) + 1(0.3) + 2(0.2) = 0.7.	**0.7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fair coin wins $3 for heads and loses $1 for tails. What is the expected value? Student's answer: 2

Use probability 1/2 for each outcome.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
E = 0.5(3) + 0.5(-1) = 1.5 - 0.5 = 1.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e ticket has a 1/1000 chance to win a $50,000 prize. What is the expected value of the prize? Student's answ

Multiply the prize by the probability.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 1/100. 50,000 × 1/1000 = 50.	**50**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the expected number of heads in 10 fair coin flips? Student's answer: 10

Multiply the number of flips by the probability of heads.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the number of flips. 10 × 1/2 = 5.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

X is 0 with probability 0.5 and 2 with probability 0.5. What is the variance of X? Student's answer: 2

Find $E(X)$, then $E(X^2) - [E(X)]^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$E(X) = 1$ and $E(X^2) = 2$. Variance $= 2 - 1^2 = 1$.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the variance of rolling a fair six-sided die? Student's answer: 3.5

Use the variance formula for a discrete random variable.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3.5 is the mean. The variance is 35/12, about 2.92.	**35/12, about 2.92**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A random variable has variance 9. What is its standard deviation? Student's answer: 9

Take the square root of the variance.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The standard deviation is $\sqrt{9} = 3$.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

If Var(X) = 3 and Y = 2X, what is Var(Y)? Student's answer: 6

Multiply the variance by the square of the constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$\mathrm{Var}(2X) = 2^2 \cdot \mathrm{Var}(X) = 4 \cdot 3 = 12.$	**12**

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Find the Error

S.MD.2

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the variance of a constant random variable? Student's answer: the same as the constant

Think about whether a constant varies.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A constant never varies, so its variance is 0.	**0**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

X is 1 or 2 with probability 0.5 each. What is the variance? Student's answer: 0.5

Find the mean, then the average squared distance.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The mean is 1.5. Each value is 0.5 from the mean, so the variance is $0.5^2 = 0.25$.	**0.25**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A ticket costs $2. It wins $6 with probability 1/4 and loses otherwise. Is this a fair game? Student says: Yes

Find the expected net value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Expected winnings are 6 × 1/4 = 1.5 and the cost is 2. The expected value is -0.5, so the game is not fair.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A game pays $10 with probability 0.1 and $0 otherwise. What is a fair price to play? Student's answer: $10

A fair price equals the expected payout.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The expected payout is 10 × 0.1 = $1, so a fair price is $1.	**$1**

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Find the Error

S.MD.3

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A game pays $5 with probability 1/2 and $0 otherwise. What is a fair price to play? Student's answer: $5

A fair price equals the expected payout.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The expected payout is 5 × 1/2 = $2.50.	**$2.50**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

abilities 0.2, 0.3, and 0.4 are assigned to three outcomes. Is this a valid probability distribution? Student says

Check that the probabilities add to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.2 + 0.3 + 0.4 = 0.9, not 1. The distribution is not valid.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

It pays a net gain of $3 with probability 0.2, and otherwise a net loss of $1. What is the expected value? Stud

Multiply each net gain or loss by its probability.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The loss was ignored. E = 0.2(3) + 0.8(-1) = 0.6 - 0.8 = -0.2.	**-0.2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A game has an expected value of 0. Is it fair? Student says: No

Think about what a fair game means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A game with expected value 0 is fair, because neither player is expected to gain over time.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

t earns $100 with probability 0.6 and loses $50 with probability 0.4. What is the expected value? Student's a

Multiply each outcome by its probability and add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student ignored the loss. E = 100(0.6) - 50(0.4) = 60 - 20 = 40.	**40**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

cted value 3 and option B has expected value 2.5. Which should be chosen to maximize expected value? S

Compare the expected values.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 is greater than 2.5, so option A has the greater expected value.	**A**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e premium is $200. The chance of a $5,000 loss is 0.02. Is the premium lower than the expected loss? Stude

Find the expected loss.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The expected loss is 5,000 × 0.02 = $100. The premium of $200 is higher than $100.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ory makes 200 items with a 3% defect rate. What is the expected number of defective items? Student's ans

Multiply the number of items by the defect rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 1.5%. 200 × 0.03 = 6.	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s 500 tickets at $10 each. One ticket wins a $2,000 prize. What is the expected value of one ticket? Student's

Subtract the ticket price from the expected prize.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the ticket cost. The expected prize is 2,000 ÷ 500 = 4, and the ticket costs $10, so the expected value is -6.	**-6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

probability 0.7 and Strategy 2 with probability 0.5. Which strategy is better for the chance of success? Stu

Compare the probabilities of success.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.7 is greater than 0.5, so Strategy 1 has the higher chance of success.	**Strategy 1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A player wins if a fair die shows an even number and loses otherwise. Is this game fair? Student says: No

Compare the probabilities of winning and losing.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Even numbers are 3 out of 6, so each side has probability 1/2. The game is fair.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e a fair spinner with 4 equal sections to choose one student for a prize. What is each student's chance? Stu

Divide 1 by the number of equal outcomes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 4 equal sections, so each student has a 1/4 chance.	**1/4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e are rolled. A player wins if the sum is 7 or more, and the other player wins otherwise. Is this fair? Student s

Count the outcomes with a sum of 7 or more.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 21 outcomes with a sum of 7 or more out of 36, which is 7/12, not 1/2. The game is not fair.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ndom number from 1 to 30 is used to choose a student. What is each student's chance? Student's answer:

Divide 1 by the number of possible numbers.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 30 numbers, so each has probability 1/30.	**1/30**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

fair spinner with 3 equal sections is used so each of 3 players has a turn. Is the method fair? Student says: N

Check whether each player has equal chance.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each player has a 1/3 chance, so the method is fair.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Player A wins if the sum of two dice is 5 or less. What is P(A wins)? Student's answer: 1/2

List the sums that are 5 or less.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sums 2, 3, 4, and 5 have 1 + 2 + 3 + 4 = 10 outcomes. P = 10/36 = 5/18, so the game is not fair.	**10/36 = 5/18**

Original task. Format inspired by error analysis routines.